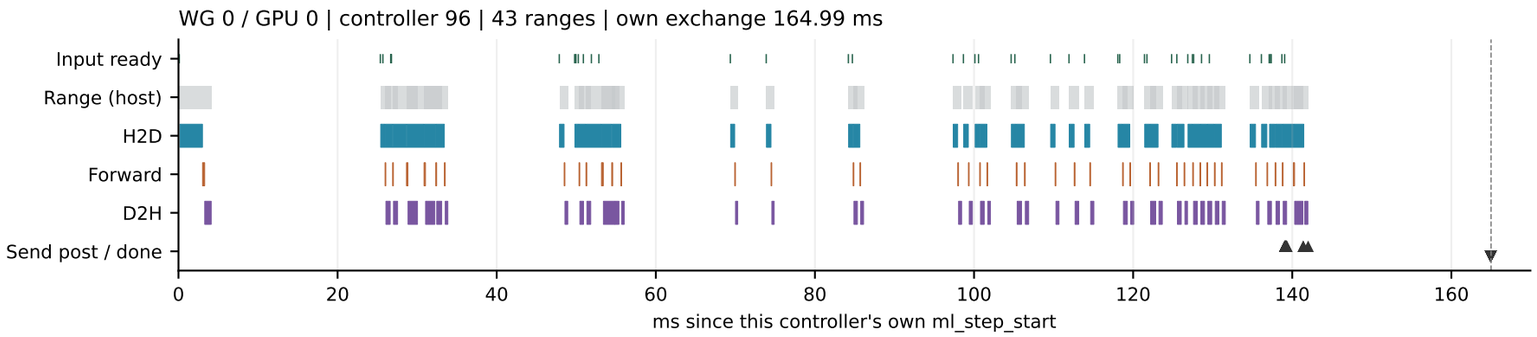
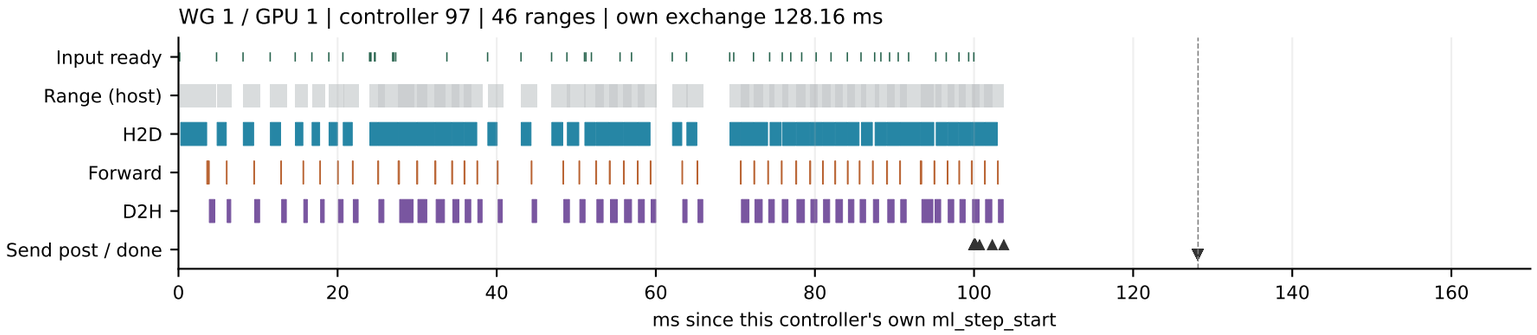
mlp | 96p96c4g | default | collective vs p2pfull | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 349.01 ms | p2pfull 227.64 ms



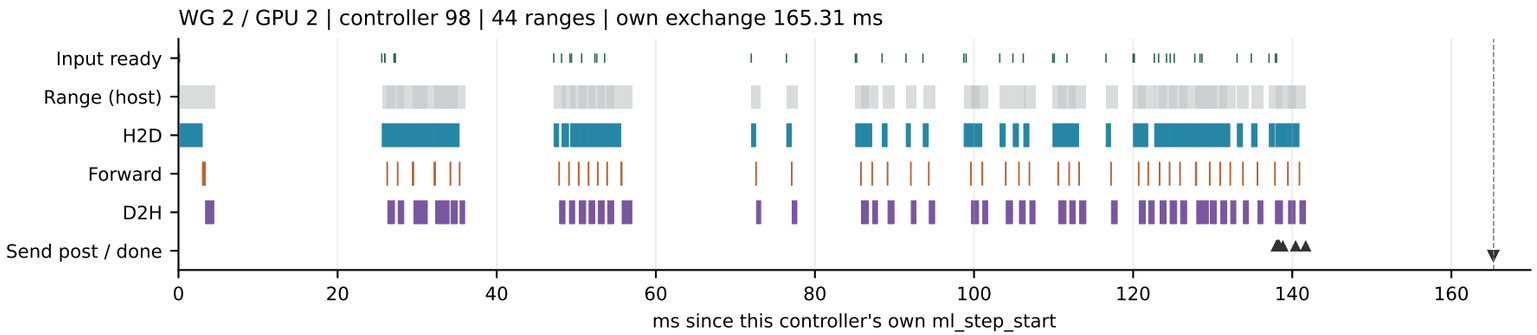
GPU work (ms)	Collective	p2pfull
H2D	25.940	39.294
Forward	2.605	4.761
D2H	12.685	22.093
GPU SUM	41.231	66.149

Controllers: coll 96 / P2P 96
Coll host device elapsed: 55.321 ms
P2P GPU envelope: 141.780 ms
Different scopes; not an elapsed-time comparison.



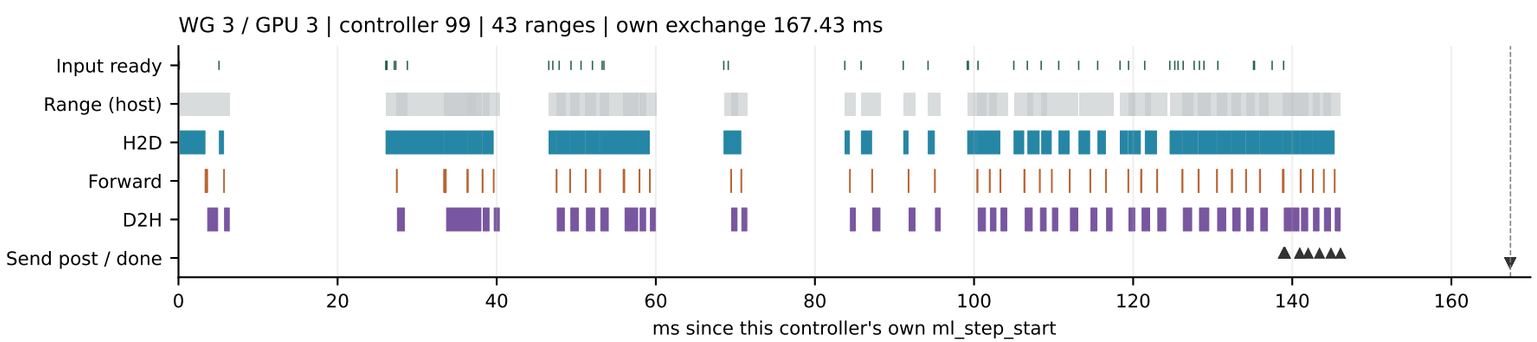
GPU work (ms)	Collective	p2pfull
H2D	49.462	71.251
Forward	2.602	5.504
D2H	19.346	34.406
GPU SUM	71.410	111.161

Controllers: coll 97 / P2P 97
Coll host device elapsed: 71.579 ms
P2P GPU envelope: 103.383 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	22.773	48.321
Forward	2.605	4.990
D2H	24.532	35.695
GPU SUM	49.911	89.006

Controllers: coll 98 / P2P 98
Coll host device elapsed: 50.055 ms
P2P GPU envelope: 141.454 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	24.425	69.331
Forward	2.601	5.719
D2H	24.322	39.881
GPU SUM	51.347	114.932

Controllers: coll 99 / P2P 99
Coll host device elapsed: 51.550 ms
P2P GPU envelope: 145.814 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.